A shared item representation forces a three-way trade-off in neural IRT

the gradient flow predicts it; the experiments confirm it

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Setup: amortized neural IRT

- Encoder (LSTM): response history $\rightarrow$ ability θ_t
- Item embedding $\rightarrow$ discrimination α_i, difficulty (step thresholds) β_i; GPCM decoder

$$P(Y_i = k \mid \theta) = \frac{\exp \sum_{m=1}^{k} \alpha_i(\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^{l} \alpha_i(\theta - \beta_{i,m})} \qquad \text{(2PL at } K{=}2\text{)}$$

- Trained on response prediction only. No supervision on θ, α, β.
- Scale is gauge-free $\Rightarrow$ recovery = rank (Spearman ρ).

The item code has two consumers

- a **value** embedding v_q (narrow) $\rightarrow$ encoder input $\rightarrow$ **ability** θ
- a **key** embedding k_q (wide) $\rightarrow$ **discrimination** α, **difficulty** β readouts

- **Shared**: one embedding, width W, feeds both consumers
- **Decoupled**: a narrow value embedding + a separate wide key embedding

$\Rightarrow$ *shared ties the encoder-input width to the readout width.* That tie is the problem.

From loss to gradient: defining z, p, y, r

One response: logit $z = \alpha(\theta - \beta)$, predicted probability $p = \sigma(z)$, observed response $y \in \{0, 1\}$.

Training loss = cross-entropy (negative log-likelihood): $L = -y \log p - (1 - y) \log(1 - p)$.
The sigmoid-with-cross-entropy (canonical-link) identity gives the **residual**

$$r := \partial L / \partial z = p - y.$$

Chain rule through z, with $\partial_\theta z = \alpha,\ \partial_\beta z = -\alpha,\ \partial_\alpha z = \theta - \beta$:

$$\partial_\theta L = r\alpha, \qquad \partial_\beta L = -r\alpha, \qquad \partial_\alpha L = r(\theta - \beta).$$

Does the loss matter? Yes. The clean $r = p - y$ is the cross-entropy/canonical-link pairing; squared error or a reweighted ordinal loss changes r. GPCM is the same form with $r_k = p_k - y_k$.

The math: α is the slow eigen-mode (derived)

From these gradients, the per-response score is one vector $s = (\alpha,\ \theta - \beta,\ -\alpha)$, so the Fisher $F = w\, ss^\top$ ($w = p(1{-}p)$) is **rank one**. The (θ, α) block has exact eigenvalues

$$\lambda_\pm = \tfrac{1}{2}\Big[(I_\theta{+}I_\alpha) \pm \sqrt{(I_\theta{-}I_\alpha)^2 + 4I_{\theta\alpha}^2}\Big], \qquad \lambda_- \approx I_\alpha(1-\rho^2).$$

- the slow mode is α: $I_\alpha = \mathbb{E}[w(\theta-\beta)^2]$ is suppressed (w peaks where $\theta{=}\beta$)
- coupling $\rho^2 = I_{\theta\alpha}^2/(I_\theta I_\alpha)$ worsens it: $\kappa_2 = I_\theta/[I_\alpha(1-\rho^2)] \geq I_\theta/I_\alpha$

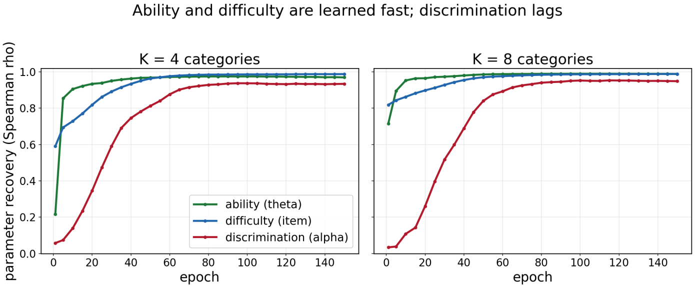
The math singles out α before any experiment.

What the math predicts (then we test each)

1. **Discrimination recovers last** (it is the single slow mode; θ, β fast).
2. **A shared code forces a three-way trade-off**: θ wants it narrow, α wants it wide, β is indifferent.
3. **Sharing adds conditioning** $\kappa = I(\theta)/I(\alpha)$: it throttles α's rate; decoupling removes it (speedup $\sim \kappa$, growing with K).
4. **Positivity is necessary, the map shape is not**: among smooth positive maps the convergence is the same; exp is not special.

Synthetic GPCM, known θ, α, β, LSTM. Each prediction is tested next.

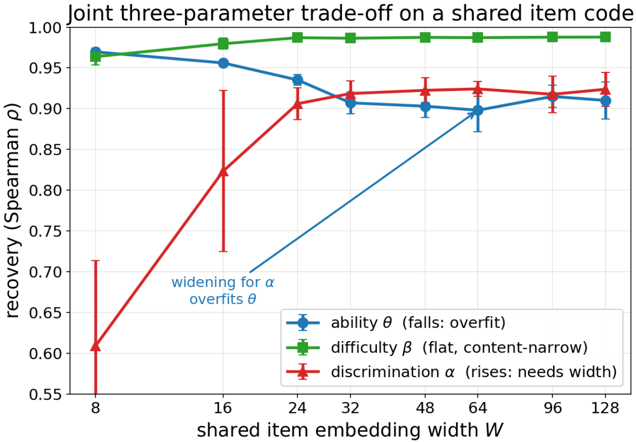
Prediction 1: discrimination recovers last



- ability, difficulty: fast (high Fisher rate)
- discrimination: slow, lags by hundreds of epochs

Exactly the ordering the rates predict.

Prediction 2: the three-way trade-off



As the shared code widens:

- α rises $0.61 \to 0.92$ (gets capacity)
- θ falls (overfits the wide input)
- β flat-high $0.96 \to 0.99$

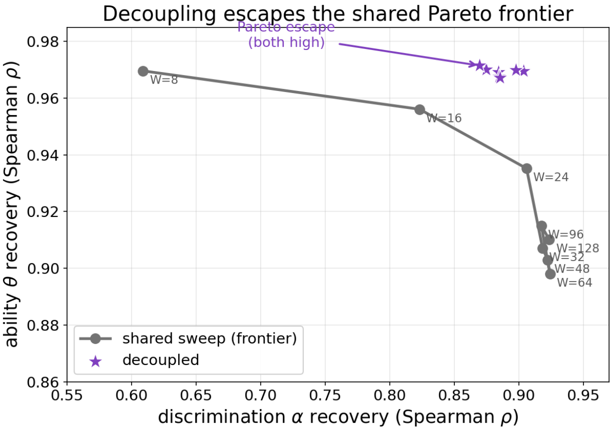
One knob, opposed pressures. β does not pay.

The two pressures are of different KIND

- α**: a RATE effect, endpoint invariant.** α is a free per-item table, so the optimum is unbiased (free-table invariant). Sharing only slows it (conditioning κ); decoupling speeds it. Transient: vanishes as training $\to \infty$, not at a fixed budget. (P9)
- θ**: an ENDPOINT effect, not invariant.** θ is read by the amortized encoder, NOT a free table. A wide input overfits it, $\mathrm{Var}(\hat{\theta}) \sim \sigma^2 \, W/n$. It does NOT vanish at convergence. (P10)
- β**: indifferent control.** High Fisher, non-pooled $\Rightarrow$ neither needs width nor is hurt ($\Delta_\beta \approx 0$). (P11)

A slow mode (rate) *and* an overfit (endpoint), tied by one width.

Prediction 3a: not capacity, allocation

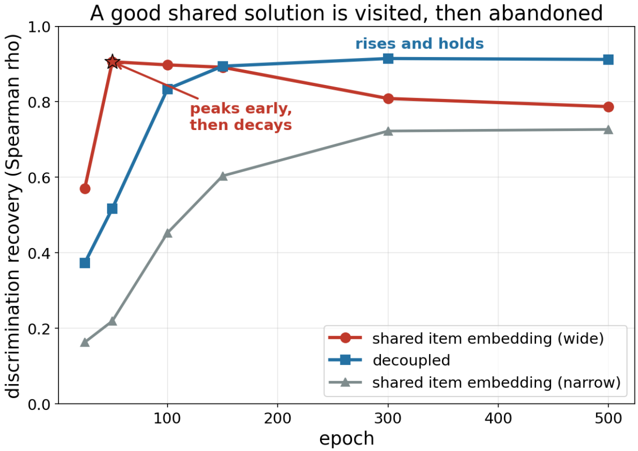


Control: matched total item-embedding params.

- widening the shared code traces a Pareto frontier
- decoupled sits **above** it, even at $6\times$ the shared budget

$\Rightarrow$ allocation, not budget. Decoupling *escapes* the frontier (P12).

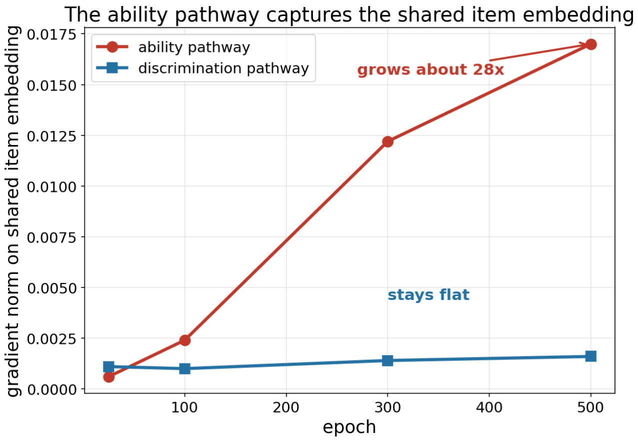
Prediction 3b: the good α solution is reached then left



- wide-shared α: peaks 0.91 (ep 50), decays to 0.79 (ep 500)
- decoupled: rises 0.91, holds

$\Rightarrow$ reachable, reached, then abandoned
= dynamics, not a ceiling.

Prediction 3c: the shared code is conditioned



- θ-pathway pull grows $\sim 28\times$; α-pathway flat
- $\cos(g_\theta, g_\alpha) \approx 0$ (not a tug-of-war)
- the lag is eigenvalue spread, stiffness $\kappa \approx 1.9$ at $K{=}4$

Prediction 3: the gradient-flow ODE

One item's embedding e_q. Near its optimum, $\delta = e_q - e_q^*$, H_q the Hessian in e_q's directions:

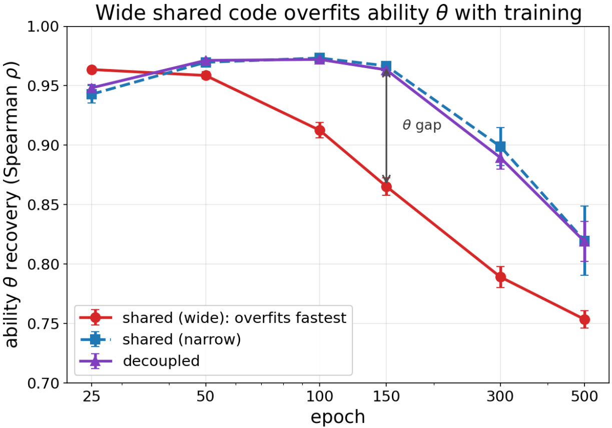
$$\dot{\delta} = -H_q \delta, \qquad \delta_c(t) = \delta_c(0) e^{-\lambda_c t}.$$

- Shared: e_q feeds encoder (θ) and α, β readouts $\Rightarrow$ steep θ direction ($\sim I(\theta)$), flat α direction ($\sim I(\alpha)$): $\kappa = I(\theta)/I(\alpha) \gg 1$
- GD, $\delta_c(t) = (1 - \eta\lambda_c)^t \delta_c(0)$, stable $\eta = 1/\lambda_{\max}$: slow mode $(1 - 1/\kappa)^t \Rightarrow t_\alpha^{\mathrm{sh}} = \kappa \log(1/\epsilon)$; decoupled $O(\log \frac{1}{\epsilon})$

$$\boxed{\text{speedup} = O(\kappa), \quad \kappa = I(\theta)/I(\alpha) \text{ (grows with } K\text{)}}$$

Same fixed point $\Rightarrow$ rate, not endpoint. Largest under GD; Adam compresses κ.

The θ side: an endpoint overfit



- wide-shared θ overfits worst: $0.96 \to 0.75$
- decoupling keeps the encoder narrow $\Rightarrow$ protects θ at the operating point

Bias and variance: $\mathrm{Var}(\hat{\theta}) \sim \sigma^2\, W/n$. An **endpoint** loss (not a rate), outside the free-table invariant. (The decay with long training, even narrow, is a separate regularization issue, not the width effect.)

The resolution: untie the two widths

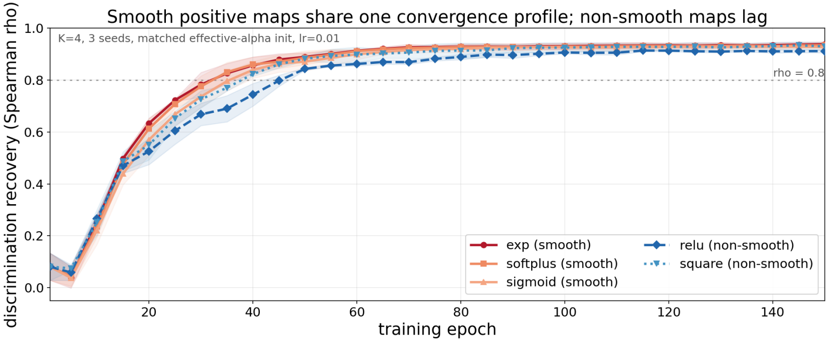
- **Shared**: one width, opposed knobs ($d\alpha/dW > 0$, $d\theta/dW < 0$, $d\beta/dW \approx 0$) $\Rightarrow$ a Pareto frontier
- **Decoupled**: two independent widths $\Rightarrow$ narrow encoder (good θ, β) $+$ wide readout (good α)

$\Rightarrow$ decoupling is Pareto-**dominant**: it escapes the frontier, not moves along it. (P12)

It cures both pressures at once: θ keeps its bottleneck, α gets its capacity.

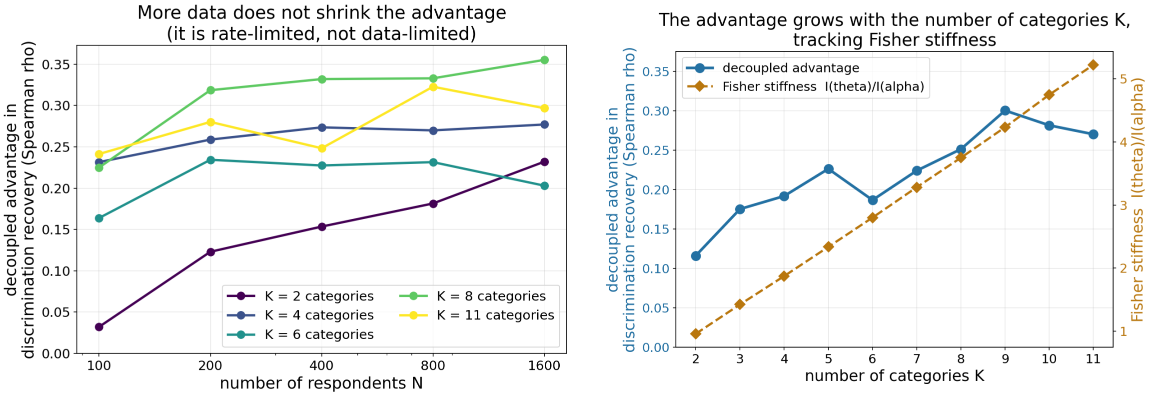
Prediction 4: positivity needed, map shape is not

Derived (Thm P7): for smooth monotone g, $m_g(\alpha) = [g'(g^{-1}(\alpha))]^2 > 0$ is a positive time-change of one canonical flow $\Rightarrow$ same fixed point and rank. exp is not special.



- confirmed: smooth maps share one profile; non-smooth lag (relu dead zone $m_g = 0$; square non-injective)
- direct-α control (θ, β frozen): a scalar reparameterization changes only speed, not the fixed point ($\kappa = 1$)

The α rate effect: finite-sample and K-scaled



- Fixed budget, vary N: the decoupling advantage does NOT shrink with data (rate-limited).
- Across K=2 to 11: it tracks the Fisher stiffness κ, $\rho = 0.89$.

What we proved (results and insights)

- **The map is a preconditioner, not a lever.** Smooth monotone positive maps share one canonical flow, so the fixed point and rank are map-invariant (Thm P7). Positivity and smoothness matter; exp does not.
- α **is the rank-one Fisher's slow direction.** $I_\alpha = \mathbb{E}[w(\theta - \beta)^2]$ is suppressed where responses concentrate; coupling makes $\kappa_2 = I_\theta/[I_\alpha(1-\rho^2)] \geq I_\theta/I_\alpha$.
- **Sharing is one ill-conditioned code block.** A single step size resolves α in $t_\alpha = \kappa \log(1/\epsilon)$ iterations; decoupling gives $O(\log \frac{1}{\epsilon})$. Speedup $O(\kappa)$, growing with K.
- **Two effects of different kind.** α a convergence-RATE effect (endpoint-invariant); θ an ENDPOINT overfit ($\mathrm{Var} \sim W/n$); β indifferent.

Conclusion

The gradient flow predicts, and the experiments confirm, that a shared item code forces a three-way trade-off: θ overfits wide, α needs wide, β is content-narrow. Decoupling unties the widths and dominates.

- Two mechanisms of different kind: α a convergence-RATE effect (endpoint invariant, $O(\kappa)$, grows with K); θ an ENDPOINT overfit ($\mathrm{Var} \sim W/n$). β indifferent.
- A measurement-validity condition: recovered parameters are otherwise representation-allocation artifacts.
- Positivity map ruled out as the lever.

Scope: synthetic GPCM, LSTM (architecture-independent on swap), 3 to 10 seeds. θ-overfit formalized for a linear amortizer (LSTM lift argued). Open: NRM slopes, GRM, real data.